

⚡ **2,558 lives**

lost to **lightning in India in 2023: 39.7%** of all deaths from forces of nature, the largest single share. Today's operational nowcasts look only **3 hours** ahead.

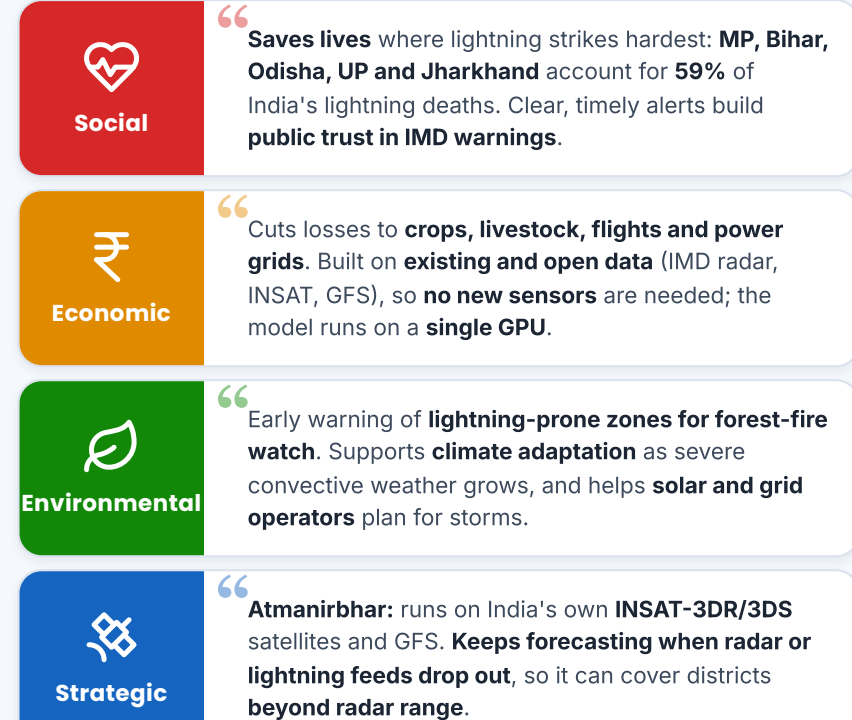
PS SIH26072 · MoES / IMD

AI/ML nowcasting of thunderstorms & lightning from radar, satellite, lightning and NWP data

Potential Impact on Target Audience



Benefits of the Solution



Measured, not claimed

Held-out 2019 storms scored against real satellite lightning (GLM).

1.8×

better than persistence at hour 1

Lightning CSI 0.613 vs 0.341; the gap widens to 5.5× by hour 3

+70%

over raw NWP lightning at hour 6

CSI 0.297 vs 0.175, and it leads NWP at every hour from 1 to 6

0.461

hour-1 CSI with no radar

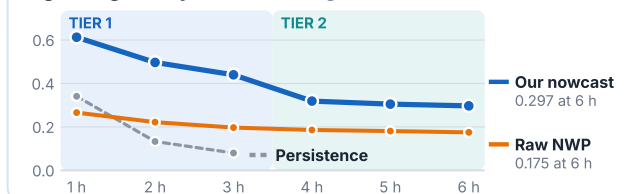
Satellite + NWP only, the usual case outside Indian radar coverage

0.71 vs 0.04

India: storm vs dry day

Storm probability 1 h ahead: Bhubaneswar outbreak (2 Sep 2023) vs Delhi dry day

Lightning CSI by lead hour (higher is better)



⚡ **“From a 3-hour to a 6-hour lightning warning: one seamless AI nowcast, built on India’s own satellites.”** ⚡

CSI = critical success index for lightning in a 16 km cell during each lead hour; Tier 1 is used for hours 1-3 and Tier 2 for hours 4-6. India probabilities from GK2A satellite + GFS input, no radar. Trained on US GLM lightning, adapted to INSAT/GFS, not yet verified against Indian lightning observations. Death figures: NCRB, Accidental Deaths & Suicides in India 2023.